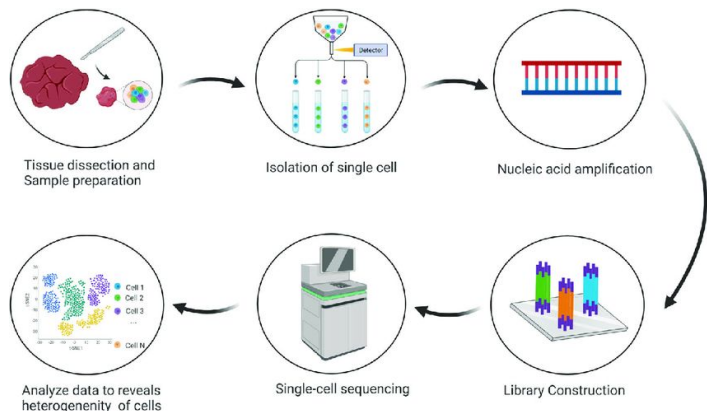
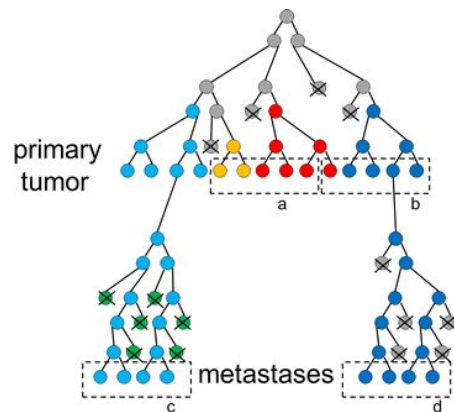


# Benchmarking Computational Methods for Cancer Cell Lineage Tracing

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# Why Study Tumor Lineages? (Biological Motivation)

- Cancer is an evolutionary process; tumors arise from a single cell and **diversify through mutations**
- Understanding **clonal evolution** helps identify treatment-resistant subpopulations and predict relapse
- Single-cell sequencing enables direct observation of heterogeneity, but **inferring evolutionary relationships** is still challenging
- Accurate lineage reconstruction reveals how driver mutations (e.g., *TP53*, *KRAS*, *PIK3CA*) shape tumor progression



Single-cell sequencing provides the raw data (mutations + expression) used to infer tumor lineage trees

Single-cell sequencing allows us to observe heterogeneity directly ...  
but accurate lineage reconstruction remains difficult due to dropout, noise, mutational sparsity, and copy-number alterations

**This motivates our benchmarking project:**

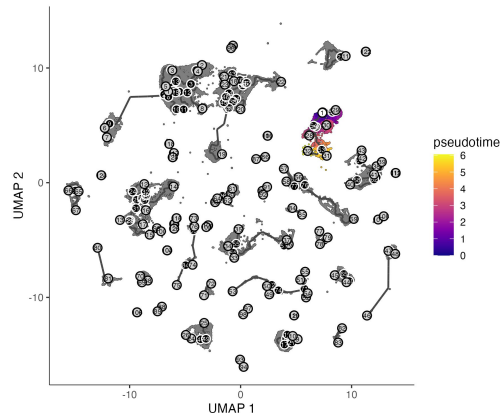
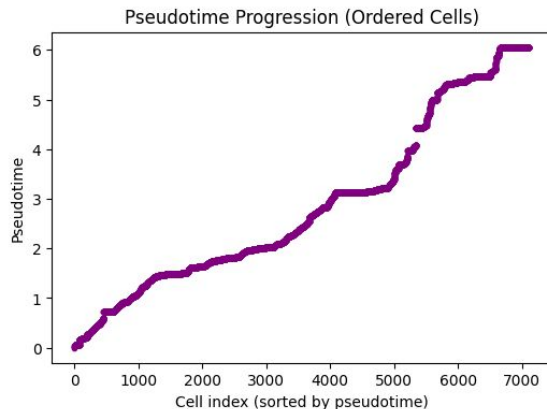
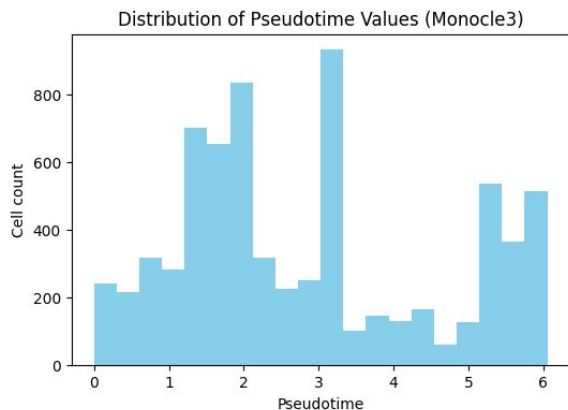
Which computational methods best reconstruct tumor evolution from single-cell data?

# Methods Overview

1. **Real data via Monocle3 (exploratory only)**
  - Used Monocle3 on a real breast-cancer scRNA-seq dataset to learn pseudotime and branching structure from expression, not from mutations
  - Monocle3 is a separate “biological motivation / context” result, not part of the synthetic benchmark
2. **Generate synthetic lineage data with ground truth**
  - Built three synthetic trees (balanced, mixed, comb) and corresponding cell × mutation matrices
  - For this presentation we focus on the balanced dataset as the clean benchmark
3. **Run mutation-based lineage methods on the synthetic data**
  - Evaluated on the balanced synthetic matrix: **SCITE, SiFit, PhISCS-BnB, Neighbor-Joining**
  - Additional methods were attempted, but did not produce stable or interpretable trees
4. **Comparison to ground truth**
  - For each method, we plotted its inferred tree and compared it visually to the balanced truth tree (depth, symmetry, major clades, over-/under-splitting)
  - Quantitative metrics (RF distance, ARI, edge precision/recall) are planned in next steps

# Monocle3 - Real Data Results

Monocle3 reveals pseudotime structure and branching trajectories in real breast cancer data, suggesting evolving subpopulations



## Monocle3 Findings:

- Reveals clear branching structure → multiple cell subpopulations
- Shows a continuous pseudotime progression → potential tumor evolution path

## Why this matters:

- Provides biological context before running lineage algorithms
- Helps validate whether inferred mutation trees match real functional structure

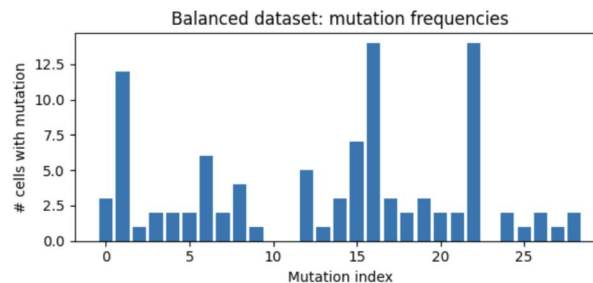
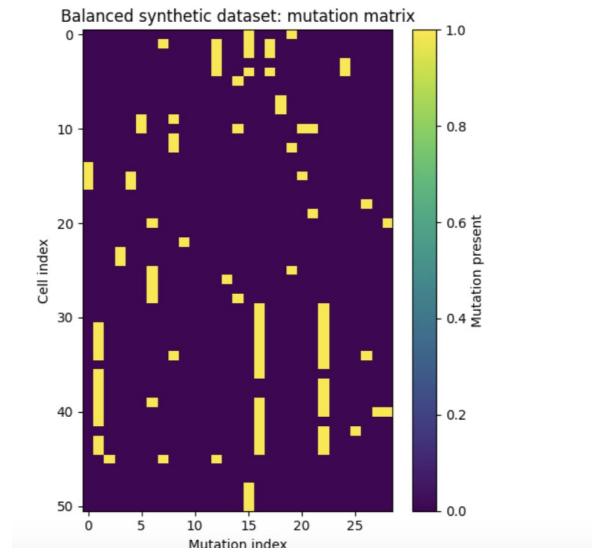
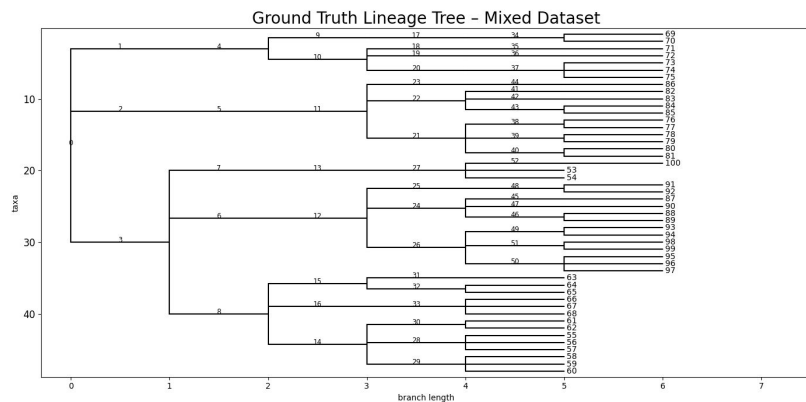
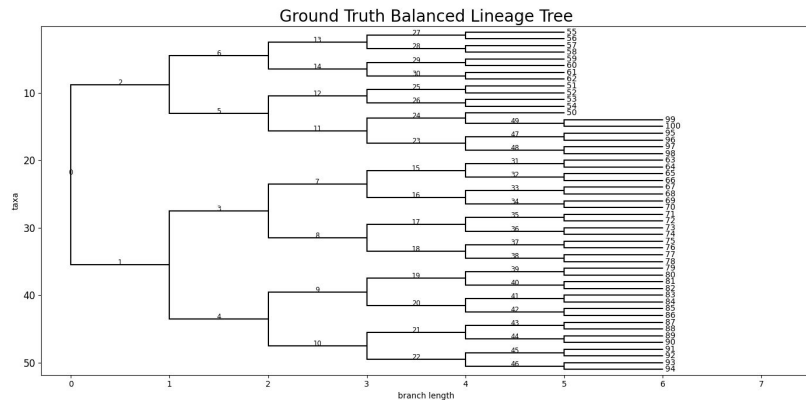
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## Process:

1. Build lineage tree - Generate a random tree structure, Three shapes: balanced (every split symmetrical), comb (uneven splitting), mixed
2. Generate cell x mutation matrix - Traverse each cell's path from root to leaf, collect all inherited mutations. Rows = cells, columns = mutations. 1 if mutation is present, 0 if not.
3. Group leaves into clones based on shared mutations
4. Apply Noise models to simulate real sequencing imperfections

# Ground-Truth Datasets for Benchmarking

Synthetic data gives us a ground truth baseline



# Neighbor-Joining

**Goal:** Reconstruct a lineage tree using a simple, fast distance-based approach.

**Inputs:**

- Binary cell  $\times$  mutation matrix
- Pairwise distance matrix derived from mutation profiles
  - (e.g., Jaccard distance on mutation presence/absence)

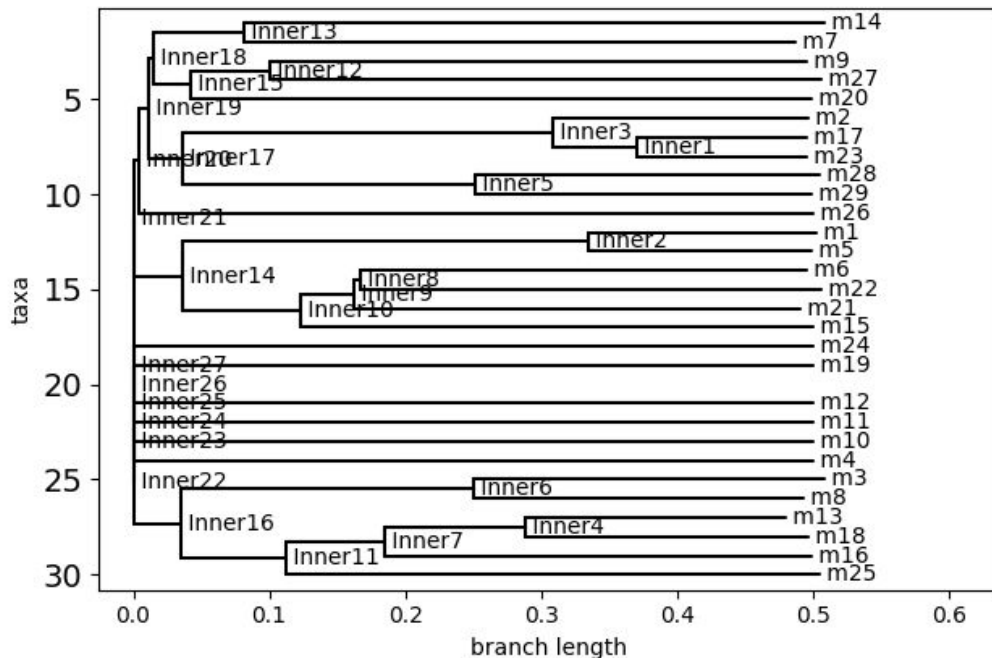
Neighbor-Joining (NJ) is a fast, distance-based phylogenetic algorithm that constructs a tree by iteratively joining the closest pairs of nodes based on mutation-derived distances. It makes no explicit evolutionary or noise assumptions and relies solely on similarity between samples.

Because of its simplicity and speed, NJ is used as a **baseline** to contextualize more sophisticated tumor-aware lineage methods. The output is a single distance-based lineage tree in Newick format.

# Neighbor-Joining Results

NJ serves as a fast, intuitive mutation-level reference tree that anchors our comparisons to more complex lineage-inference tools

## Neighbor-Joining Tree (Mutation-Level, m1-m29)



**Dataset:** balanced synthetic dataset (51 cells, 29 mutations)

**Method:**

- Input: binary mutation matrix
- Distance metric: Jaccard distance between mutations
- Output: mutation-level tree (labels: m1–m29)

**Runtime:** < 1 seconds

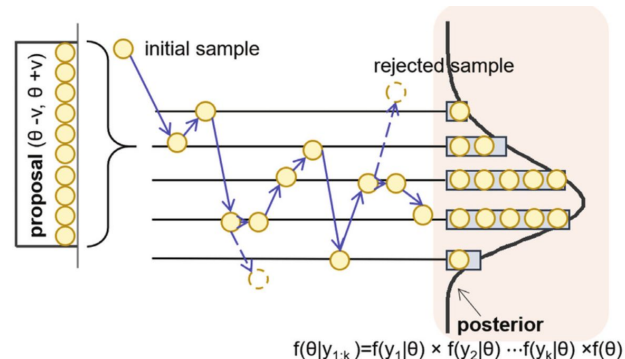


# SCITE

**Goal:** Reconstruct cell lineage trees from noisy single-cell mutation data.

**Inputs:**

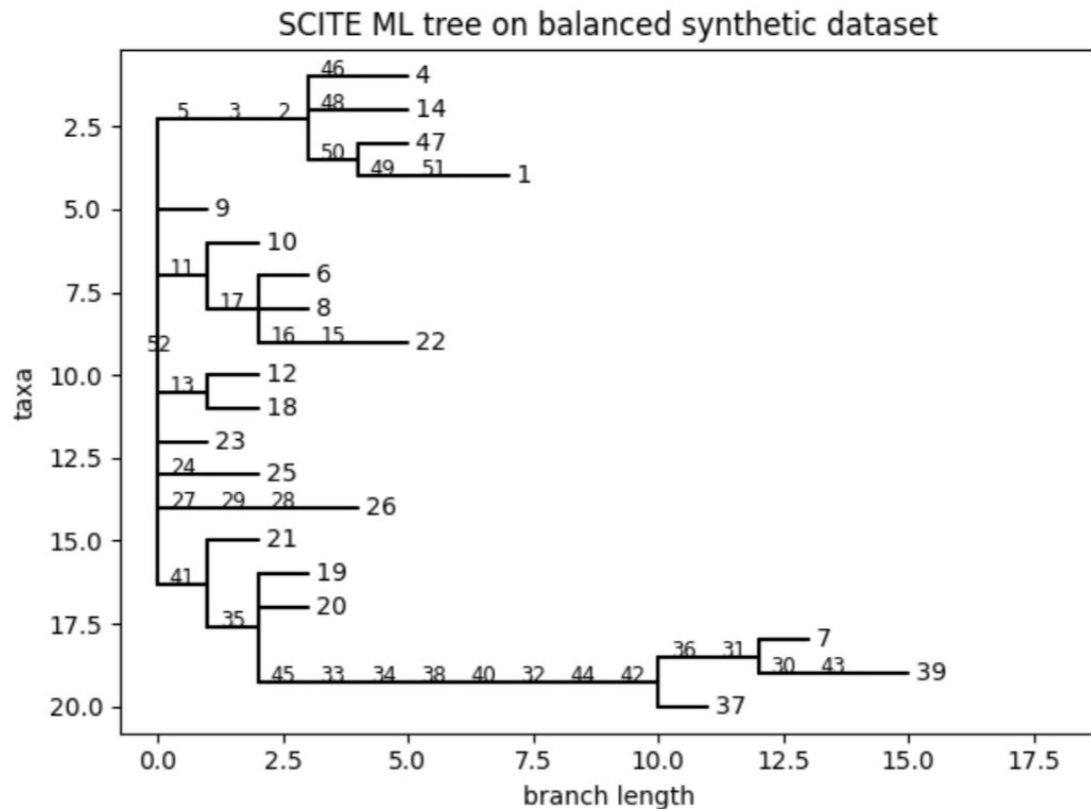
- Binary cell  $\times$  mutation matrix
- Estimated sequencing error rates:
  - $\alpha$  = false positive rate
  - $\beta$  = allelic dropout / false negative rate



SCITE formulates cell lineage reconstruction as a **probabilistic inference problem** over mutation events and tree structures. Given a binary cell-by-mutation matrix, it uses **maximum likelihood estimation with MCMC** sampling to infer mutation order and lineage topology while explicitly modeling **single-cell sequencing** errors such as false positives and allelic dropout. The method outputs **maximum-likelihood lineage trees** in Newick format along with **posterior samples** over mutation histories, enabling uncertainty assessment, performance benchmarking, and downstream comparison to ground-truth lineages.

# SCITE Results

SCITE recovers major branches but may miss fine structure



**Dataset:** balanced synthetic dataset (51 cells, 29 mutations)

**SCITE settings:**

- False negative rate  $\beta = 0.10$
- False positive rate  $\alpha = 0.05$
- Maximum likelihood (ML) tree from SCITE

**Runtime:** ~193 seconds on this dataset

**Interpretation:**

- SCITE recovers a branching clonal structure consistent with the “balanced” design of the synthetic tree
- This gives us a baseline lineage for comparison against PhyloWGS and other methods

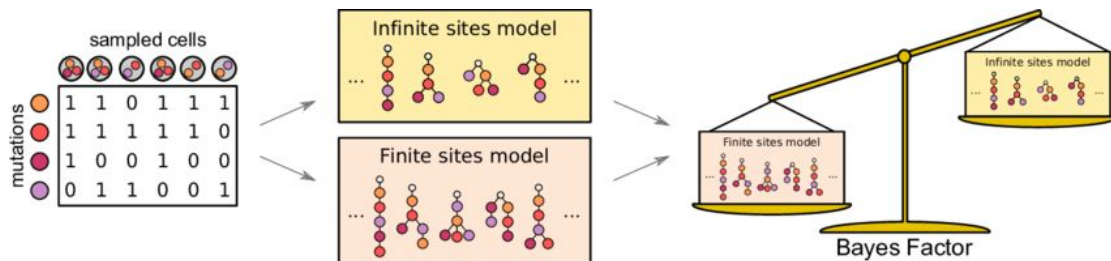
# Phylogeny Inferred Using SiFit (Finite-Sites Model)

**Goal:** Reconstruct tumor lineage trees from noisy single-cell mutation data under a biologically realistic mutation model.

SiFit reconstructs tumor lineages using a **maximum-likelihood framework** under a finite-sites mutation model, allowing mutations to recur or be lost while explicitly modeling sequencing errors. By relaxing the infinite-sites assumption made by tools like SCITE, SiFit captures more biologically realistic mutation processes, often resulting in **deeper and more complex lineage trees**.

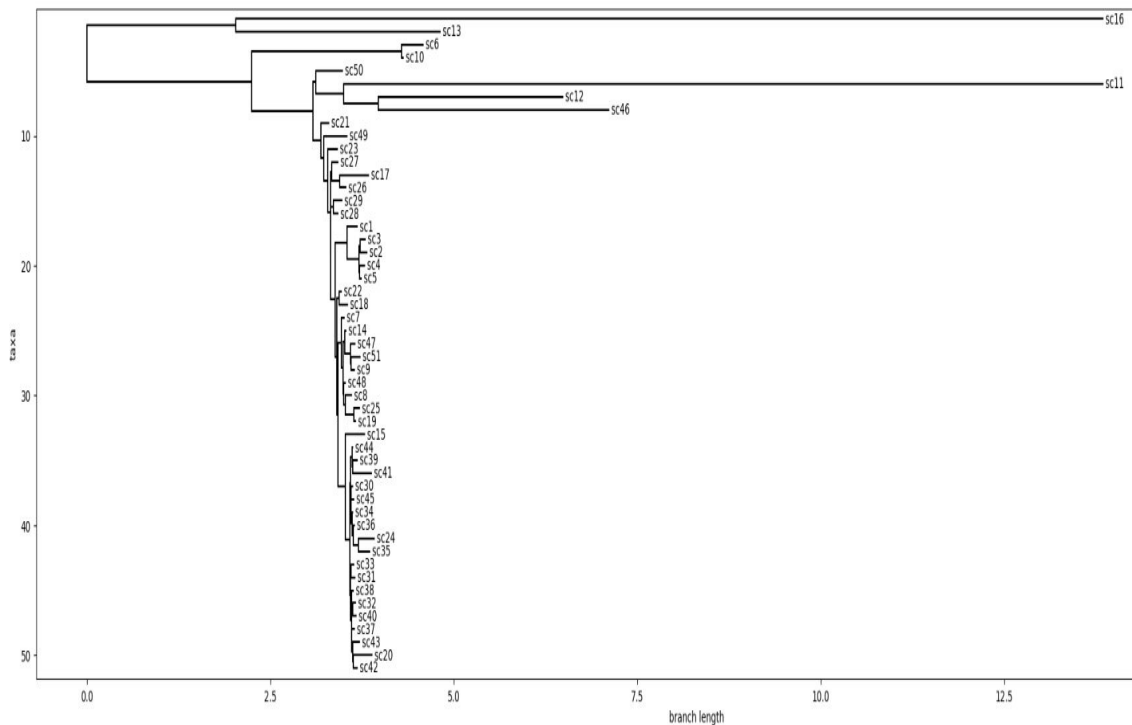
## Inputs:

- Binary cell  $\times$  mutation matrix
- Estimated sequencing error parameters
  - False positive rate
  - False negative (dropout) rate



# SiFit Results

SiFit offers a finite-sites perspective on the same dataset, revealing mutational patterns that infinite-sites tools may miss



**Dataset:** balanced synthetic dataset (51 cells, 29 mutations)

**SiFit settings:**

- False negative rate (FN) = 0.10
- False positive rate (FP) = 0.01
- Finite-sites evolutionary model
- 10,000 iterations (maximum likelihood search)
- Binary mode (-df 0, required for SC matrices)

**Output:** Maximum likelihood mutation tree

**Runtime:** ~7 seconds

**Interpretation:**

- SiFit reveals deeper, more uneven branching consistent with real tumor subclonal diversity rather than the clean symmetry seen in synthetic data
- Long isolated branches highlight heterogeneous mutation burdens, suggesting biologically distinct subpopulations rather than uniform clonal expansion.

# PhISCS-BnB

**Goal:** Infer a clonal lineage tree that satisfies perfect-phylogeny constraints from single-cell mutation data.

**Inputs:** Binary cell  $\times$  mutation matrix (cells  $\times$  mutations)

PhISCS-BnB reconstructs tumor phylogenies by formulating lineage inference as an **integer linear programming (ILP)** problem solved via **branch-and-bound optimization**. It enforces a **perfect-phylogeny model**, where each mutation arises once and is never lost, and finds an optimal solution under these constraints rather than relying on probabilistic sampling. The method outputs a **clonal lineage tree in Newick format**, optionally with mutations annotated on branches, but can be sensitive to noise and sparse mutation signals due to its strict phylogenetic assumptions.

## Algorithm 1 PhISCS-BnB

**Input:**  $I \in \{0, 1\}^{n \times m}$ : Original input to the algorithm

**Output:**  $Y \in \{0, 1\}^{n \times m}$  such that

$Y = \operatorname{argmin}_{\text{PP}(Y)} F_{0 \rightarrow 1}(I, Y)$ .

1:  $\text{Best}_{\text{Node}} \leftarrow$  A simple PP solution (see Section 3.3)

2:  $Q \leftarrow$  An empty priority queue

3: Push  $I$  in  $Q$

4: **while**  $Q$  is not empty **do**

5:    $C \leftarrow$  pop next node from  $Q$  with the lowest priority score

6:    $\text{New}_{\text{node}_1} \leftarrow C$

7:   **for**  $r \in R_{0,1}(p, q)$  **do**  $\triangleright$  See Lemma 3.1

8:      $\text{New}_{\text{node}_1}(r, p) \leftarrow 1$   $\triangleright F_{0 \rightarrow 1}(I, \text{New}_{\text{node}_1})$  is also updated here

9:    $\text{New}_{\text{node}_2} \leftarrow C$

10:   **for**  $r \in R_{1,0}(p, q)$  **do**

11:      $\text{New}_{\text{node}_2}(r, q) \leftarrow 1$

12:   **for**  $i \in \{1, 2\}$  **do**

13:     **if**  $\text{New}_{\text{node}_i}$  is a PP **then**  $\triangleright$  i.e.  $\text{New}_{\text{node}_i}$  is a leaf

14:       **if**  $F_{0 \rightarrow 1}(I, \text{New}_{\text{node}_i}) < F_{0 \rightarrow 1}(I, \text{Best}_{\text{Node}})$  **then**

15:          $\text{Best}_{\text{Node}} \leftarrow \text{New}_{\text{node}_i}$

16:     **else**  $\triangleright$  Use any bounding algorithm proposed in Section 3.2

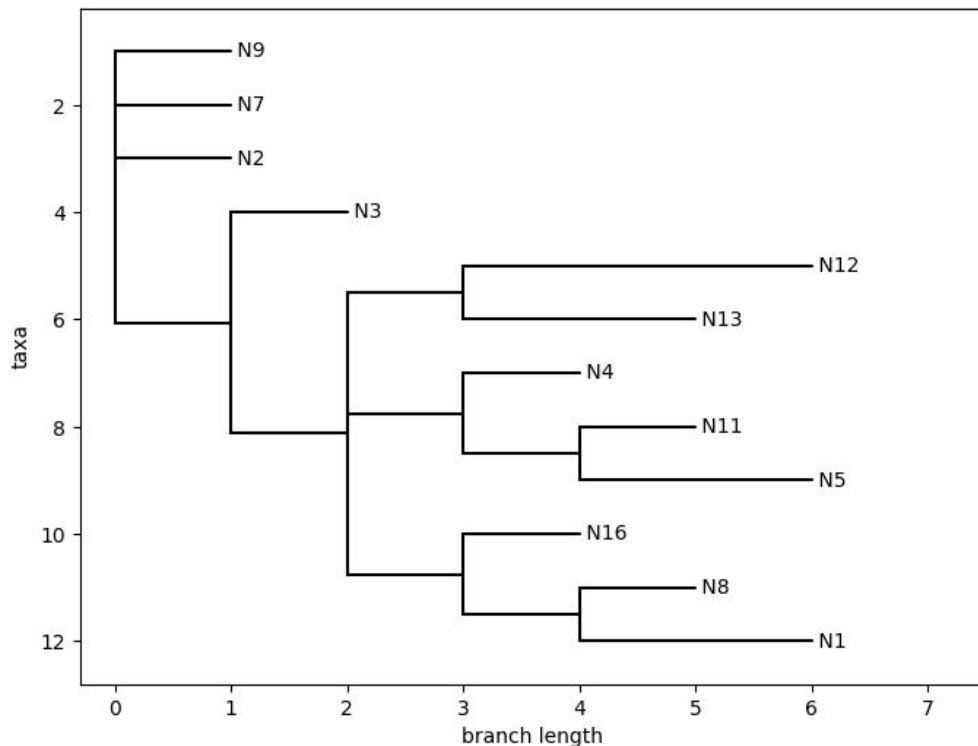
17:        $\text{lb} \leftarrow$  A lower bound for the number of flips to get to a PP matrix from  $\text{New}_{\text{node}_i}$

18:       **if**  $\text{lb} + F_{0 \rightarrow 1}(I, \text{New}_{\text{node}_i}) < F_{0 \rightarrow 1}(I, \text{Best}_{\text{Node}})$  **then**

19:         Push  $\text{New}_{\text{node}_i}$  in  $Q$  with priority score set to  $\text{lb} + F_{0 \rightarrow 1}(I, \text{New}_{\text{node}_i})$

20: **Return**  $\text{Best}_{\text{Node}}$

# PhISCS-BnB Results



**Dataset:** balanced synthetic dataset (51 cells, 29 mutations)

**SCITE settings:**

- Branching factor  $b = 2$
- Draw tree  $t = 1$

**Runtime:** ~53 seconds on this dataset

**Interpretation:**

- Unlike SCITE, requires every mutation occurs once and does not revert - causes shallow tree, limited branching
- Over-simplify lineage structure when the dataset contains noise

# Comparison of Mutation-Based Lineage Inference Methods

They all start from the same single-cell mutation matrix, but they make *very different assumptions*, which is why their trees diverge

| Feature                 | SCITE                                                                  | SiFit                                                              | PhISCS-BnB                                                    | Neighbor-Joining                                    |
|-------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------|
| Primary data            | Single-cell mutation matrix                                            | Single-cell mutation matrix                                        | Single-cell mutation matrix                                   | Mutation-derived distances                          |
| Evolutionary model      | Infinite sites                                                         | Finite sites (allows repeated mutations)                           | Perfect phylogeny (ILP)                                       | None (distance-based)                               |
| Handles dropout/noise   | Yes ( $\alpha$ , $\beta$ model)                                        | Yes (explicit error model)                                         | Limited (constraint-based)                                    | No explicit noise modeling                          |
| Output structure        | Single ML cell lineage tree                                            | Single full mutation tree                                          | Clonal tree (mutations on edges)                              | Tree based on similarity                            |
| Computational cost      | Moderate (MCMC)                                                        | Fast                                                               | Moderate                                                      | Very fast                                           |
| Strength                | Robust to scRNA-seq noise                                              | Biologically realistic mutation model                              | Guarantees phylogenetic consistency                           | Simple, assumption-free baseline                    |
| Limitation              | Infinite-sites assumption                                              | Can produce deep/complex trees                                     | Sensitive to noise/constraints                                | Not tumor-aware                                     |
| Observed in our project | Recovers some balanced structure but collapses branches under conflict | Produces rich structure on real tumor data; captures heterogeneity | Over-simplifies synthetic balanced tree into few large clones | Mismatches ground truth; non-biological bushy shape |

# Methods Not Aligned to Our Synthetic Setup

## PhyloWGS:

- Designed for bulk sequencing VAFs
- Our binary single-cell genotypes violate its assumptions → flat posterior, no consensus tree

## LinTIMaT:

- Requires mutation + expression matrices
- Mutation-only data breaks its model → cannot infer trajectories

## OncoNEM:

- Assumes infinite-sites and low noise
- Our noisy mutation model makes it return oversimplified trees

Exception: 85% of trees are multiprimary (1700 of 2000), so not enough to report good posterior.



## Next Steps

- Apply SCITE, SiFit, PhISCS-BnB, and Neighbor-Joining to a **real single-cell tumor mutation dataset**, verify conclusions from synthetic datasets
- Compare real-data lineage structures to synthetic benchmark behavior, create **numeric ranking method** to compare trees
- **Aggregate results** and draw biological interpretation running all four methods in a single real dataset
- Summarize tradeoffs to **provide guidance on method selection** for different data types

# Citations

- Jahn et al., SCITE: Inferring tumor evolution from single-cell sequencing, Bioinformatics (2016)
- Morrissy et al., Single-cell characterization of medulloblastoma reveals therapy-induced stemness, Nature (2024)
- Satas & Raphaël, Accurate, fast, and memory-efficient inference of single-cell phylogenies, Genome Research (2020)
- Ogawa et al., Computer-assisted DNA phylogeny reconstruction based on restriction enzyme cleavage patterns, Gene (1987)